

Byzantine General's Problem. It enables nodes to decide whether to accept or deny the submitted information. The party maintains an internal state which is used to run computation on new messages. If the party's computation runs well, it then decides to share the transaction with other parties within the same network. The final result is arrived at when multiple parties have run computations on it and the decision is based on the results sent by the parties. As all nodes are not required for the consensus method, it also has a low hash rate requirement. However, to make it successful, it requires good trusted nodes to work together. PBFT is used in popular projects like Ripple, Stellar and Hyperledger.

Blockchain use cases

Taking a closer look at a few blockchain use cases, from the hundreds out there, will help you understand blockchain technology better.

Supply chain management: Supply chain management (SCM) currently suffers from a lot of problems. For example, the biggest problem is product counterfeiting. By using blockchain, products can be traced accurately, making counterfeiting impossible. It also improves transparency and ensures that there is an overall cost reduction. Improvements in SCM also means improvements in different industries like healthcare, food, etc.

The Food Trust project by IBM, run in collaboration with Walmart, is an excellent example of supply chain improvement with blockchain in the food industry.

Healthcare: Healthcare also benefits from blockchain. Right now, there is no way for healthcare specialists to look at one single version of a person's healthcare report. With blockchain, it now becomes possible for patients to have their reports stored on a blockchain — which can be retrieved later on. In this way, doctors can work on a single report and give better medical services to patients. It also makes the process efficient, improves the exchange of information, and results in the best possible way for patients to both secure and access their documents so that they do not have to be carried around.

MediBlock is one of the projects that is using blockchain technology to improve healthcare services.

The energy market: Energy distribution companies are big players that provide services to almost everyone. For now, as a consumer, you need to wait for the big companies to provide you with a connection — which can take anywhere between a few days to a few weeks, depending on your location. The energy market is currently a closed ecosystem with new players trying to revolutionise it with the help of blockchain technology. With blockchain, nodes will become capable of generating and trading electricity without the need for any centralised authority.

This will reduce energy prices and also improve the reach of energy

in places where it was unavailable earlier. There are projects like Grid+ and Power Ledger that are making all of this possible.

Real estate: Real estate is one big sector. Right now, many players in the sector are struggling due to one reason or another. One major reason behind the slow growth of the real estate market is the lack of efficiency in the whole system. If you wish to purchase a property, then you need to go through a slow, painful process that can take weeks to complete.

With blockchain, real estate can become more efficient, enabling safe and smooth transactions. All of this is possible because of smart contracts that can automate tasks like buying or selling a property. It can also help carry out ownership verification and is cost-effective.

Blockchain development for beginners

If you want to get started with blockchain development, then you need to begin with programming first. It is wise to learn C++, followed by Python, JavaScript, Solidity and Go. When it comes to the choice of the blockchain network, you should start with Ethereum. It has an excellent community, and you will surely feel at home while learning it. **END** 

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